

Docker Compose Based Multi-Container Web Application with Redis Persistence

```
Command Prompt
D:\redis-devops>type docker-compose.yml
services:
  frontend:
    build: ./frontend
    container_name: frontend
    ports:
      - "3000:80"
    restart: unless-stopped
    depends_on:
      - backend

  backend:
    build: ./backend
    container_name: backend
    ports:
      - "5000:5000"
    restart: unless-stopped
    environment:
      REDIS_HOST: redis
      REDIS_PORT: 6379
    depends_on:
      - redis

  redis:
    image: redis:7-alpine
    container_name: redis
    ports:
      - "6379:6379"
    volumes:
      - redis-data:/data
    restart: unless-stopped
    command: redis-server --appendonly yes

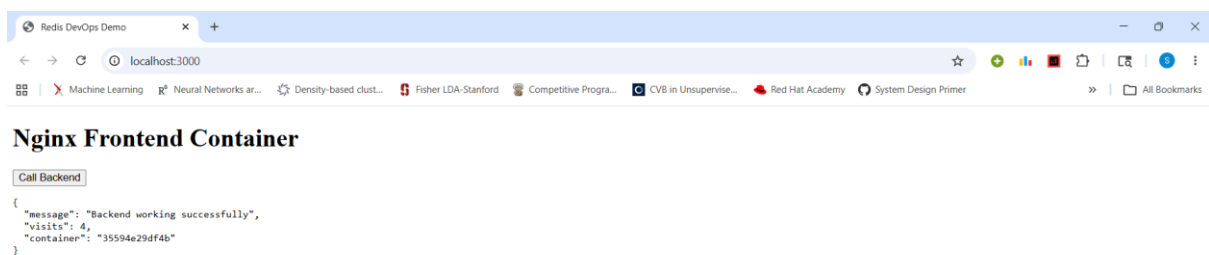
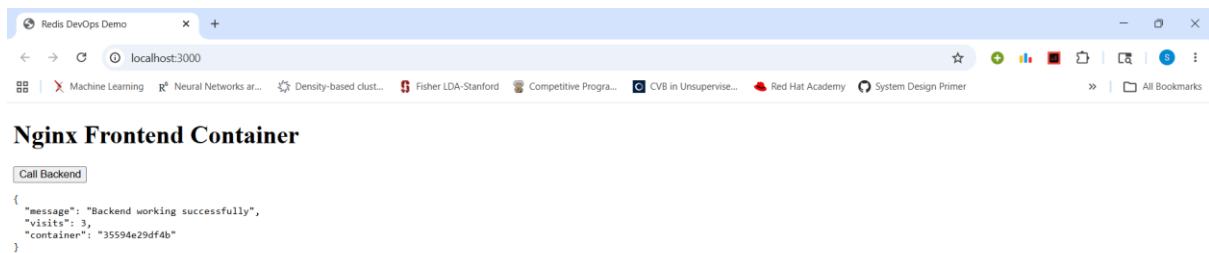
volumes:
  redis-data:
```

```
D:\redis-devops>tree /F
Folder PATH listing for volume New Volume
Volume serial number is 00000077 AEE6:4A19
D:
|
|   docker-compose.yml
|
|   -- backend
|   |   dockerfile
|   |   package.json
|   |   server.js
|
|   -- frontend
|   |   dockerfile
|   |   nginx.conf
|
|   -- public
|   |   index.html
```

```
D:\redis-devops>type backend\dockerfile
FROM node:18-alpine
WORKDIR /app
COPY package.json .
RUN npm install
COPY . .
EXPOSE 5000
CMD ["npm", "start"]
D:\redis-devops>type frontend\dockerfile
FROM nginx:latest
COPY public /usr/share/nginx/html
COPY nginx.conf /etc/nginx/nginx.conf
EXPOSE 80
D:\redis-devops>
```

```
Command Prompt - docker c x + v
#20 exporting manifest sha256:21161549bd27dcc8684fe66e7134799aaflaalfda78899949f2e137ace80b10a9 done
#20 exporting config sha256:86a29885dcb883ce063bb17f7135ba381cd7bb714f14063b00275c32e8d81afe done
#20 exporting attestation manifest sha256:45319f044c364f0e4ef0aea8ab9d1931d405b391637e737333e132217dd29f7 0.0s done
#20 exporting manifest list sha256:4d7d46694a3edac0e0f537ee96656b84464ea16252c3c7d1becfa5d5c63332 0.0s done
#20 naming to docker.io/library/redis-devops-backend:latest
#20 naming to docker.io/library/redis-devops-backend:latest done
#20 unpacking to docker.io/library/redis-devops-backend:latest 0.0s done
#20 DONE 0.2s

#21 [backend] resolving provenance for metadata file
#21 DONE 0.0s
[+] up 7/7
✓Image redis-devops-frontend Built 2.4s
✓Image redis-devops-backend Built 2.4s
✓Network redis-devops-default Created 0.1s
✓Volume redis-devops-redis-data Created 0.0s
✓Container redis Created 0.2s
✓Container backend Created 0.1s
✓Container frontend Created 0.2s
Attaching to backend, frontend, redis
redis 1:C 14 May 2026 11:23:04.232 * o000o000o000o Redis is starting o000o000o000o
redis 1:C 14 May 2026 11:23:04.232 * Redis version=7.4.9, bits=64, commit=00000000, modified=0, pid=1, just started
redis 1:C 14 May 2026 11:23:04.232 * Configuration loaded
redis 1:M 14 May 2026 11:23:04.233 * monotonic clock: POSIX clock_gettime
redis 1:M 14 May 2026 11:23:04.235 * Running mode=standalone, port=6379.
redis 1:M 14 May 2026 11:23:04.236 * Server initialized
redis 1:M 14 May 2026 11:23:04.241 * Creating AOF base file appendonly.aof.1.base.rdb on server start
redis 1:M 14 May 2026 11:23:04.248 * Creating AOF incr file appendonly.aof.1.incr.aof on server start
redis 1:M 14 May 2026 11:23:04.248 * Ready to accept connections tcp
backend > backend@1.0.0 start
backend > node server.js
frontend /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
frontend /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
frontend /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
frontend 10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
frontend 10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
frontend /docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
frontend /docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
frontend /docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
frontend /docker-entrypoint.sh: Configuration complete; ready for start up
backend Connected to Redis
backend Backend running on port 5000
```



The above two images are before container stop

```
D:\redis-devops>docker compose down
[+] down 4/4
✓Container frontend      Removed      0.1s
✓Container backend      Removed      0.1s
✓Container redis         Removed      0.1s
✓Network redis-devops_default Removed      0.3s
D:\redis-devops>
```

Container stop

```
D:\redis-devops>docker compose up
[+] up 4/4
✓Network redis-devops_default Created      0.0s
✓Container redis         Created      0.2s
✓Container backend      Created      0.2s
✓Container frontend      Created      0.2s
Attaching to backend, frontend, redis
redis | 1:C 14 May 2026 11:26:45.769 * o000o00o000o Redis is starting o000o00o000o
redis | 1:C 14 May 2026 11:26:45.769 * Redis version=7.4.9, bits=64, commit=00000000, modified=0, pid=1, just started
redis | 1:C 14 May 2026 11:26:45.769 * Configuration loaded
redis | 1:M 14 May 2026 11:26:45.771 * monotonic clock: POSIX clock_gettime
redis | 1:M 14 May 2026 11:26:45.776 * Running mode=standalone, port=6379.
redis | 1:M 14 May 2026 11:26:45.778 * Server initialized
redis | 1:M 14 May 2026 11:26:45.779 * Reading RDB base file on AOF loading...
redis | 1:M 14 May 2026 11:26:45.779 * Loading RDB produced by version 7.4.9
redis | 1:M 14 May 2026 11:26:45.779 * RDB age 221 seconds
redis | 1:M 14 May 2026 11:26:45.779 * RDB memory usage when created 0.90 Mb
redis | 1:M 14 May 2026 11:26:45.779 * RDB is base AOF
redis | 1:M 14 May 2026 11:26:45.779 * Done loading RDB, keys loaded: 0, keys expired: 0.
redis | 1:M 14 May 2026 11:26:45.780 * DB loaded from base file appendonly.aof.1.base.rdb: 0.001 seconds
redis | 1:M 14 May 2026 11:26:45.781 * DB loaded from incr file appendonly.aof.1.incr.aof: 0.001 seconds
redis | 1:M 14 May 2026 11:26:45.781 * DB loaded from append only file: 0.003 seconds
redis | 1:M 14 May 2026 11:26:45.781 * Opening AOF incr file appendonly.aof.1.incr.aof on server start
redis | 1:M 14 May 2026 11:26:45.781 * Ready to accept connections tcp
backend | > backend@1.0.0 start
backend | > node server.js
frontend | /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
frontend | /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
frontend | /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
frontend | 10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
frontend | 10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
frontend | /docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
frontend | /docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
frontend | /docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
frontend | /docker-entrypoint.sh: Configuration complete; ready for start up
backend | Connected to Redis
backend | Backend running on port 5000
```



Nginx Frontend Container

```
Call Backend

{
  "message": "Backend working successfully",
  "visits": 5,
  "container": "7e0011b663c7"
}
```

Volume persistence in action thanks to docker volumes and startup command --appendonly yes

